

Bombardier Beetle Optimizer: A Novel Bio-Inspired Algorithm for Global Optimization

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Abstract—In this paper, a novel bio-inspired optimization algorithm is proposed, called “Bombardier Beetle Optimizer (BBO)”. This type of species is very intelligent, which has the ability to defend and escape from predators. The principles of the former one is inspired by the defense mechanism of the Bombardier Beetle against the predators, which the Bombardier Beetle triggers a toxic chemical spray when it feels threatened. This reaction occurs in a specialized reaction chamber inside its abdomen and includes a well-regulated enzymatic mechanism, which comprises hot water vapor, oxygen, and irritating substances like p-benzoquinones. In addition, the proposed BBO also simulates the escape mechanism of the Bombardier Beetle from a predator, which it has the ability to calculate its distance from the predator and it can fly away. The BBO is tested with optimizing six benchmark functions of “Congress on Evolutionary Computation (CEC 2017)” test bed suites. Moreover, it is compared against well-known metaheuristic optimization algorithms, including “Chernobyl Disaster Optimizer (CDO)”, “Grey Wolf Optimizer (GWO)”, “Particle Swarm Optimization (PSO)”, “Bermuda Triangle Optimizer (BTO)”, “Sperm Swarm Optimization (SSO)” and “Gravitational Search Algorithm (GSA)”. The outcomes of this paper prove the BBO’s efficiency in which outperforms the other algorithms in terms of convergence rate and quality of results.

Keywords—Randomness, Physical-based algorithms, Swarm-based algorithms, Bio-inspired algorithms, (CEC) 2017 Benchmarks functions

I. INTRODUCTION

Real life problems have been increased with increase life demands. Example of these problems are Wireless Sensor Network (WSN) problem, biomedical problems, engineering design problems, etc. These problems can be converted to mathematical models, however, it is a challenging task to solve these problems efficiently, especially when they have a set of local minima in their geometry [1].

To tackle these problems, we rely on optimization algorithms, which are systematic methods and high level procedures designed to search for optimal or near-optimal solutions [2, 3]. These algorithms are inspired by diverse natural and artificial systems, such as biology, social behavior, physics and even historical events, which are making them versatile and robust for real-world applications. These algorithms are broadly

categorized into four main classes, including, physical based class, swarm and bioinspired based class, evolutionary based class, and manmade based class. Examples of physical based class are Chernobyl Disaster Optimizer (CDO) [4] and Bermuda Triangle Optimizer (BTO) [5]. Examples of swarm based and bioinspired class are Particle Swarm Optimization (PSO) [6], Grey Wolf Optimizer (GWO) [7] and Sperm Swarm Optimization (SSO) [8,9]. An example of evolutionary based class is Genetics Algorithm (GA) [10,11]. Examples of manmade class are Five Phases Algorithm (FFA) [12] and Fireworks Algorithm (FA) [13].

The advantages of the GSA, SSO, PSO, CDO, BTO, and GWO algorithms can be listed as follows [4-9]:

- SSO and PSO are simple to utilize and understand.
- Usually, PSO simply required to be tuned for the social coefficient, inertia weight, and cognitive coefficient. This simplicity in adjusting of parameters helps to mitigate error appearance in code.
- SSO uses random parameters in its operation, such as temperature value, and pH value, which are an exact values of vital signs of female reproduction system. To mitigate gaps between these tuning parameters, SSO utilizes the logarithm to normalize them and helps to reduce error appearance in code.
- PSO and SSO converge faster in various kinds of problems.
- SSO, GWO and PSO can be utilized to solve different kinds of optimization problems, such as both discrete and continuous problems, solving multiple objectives optimization problems, and solving problems without and with constraints.
- SSO's adaptive strategies provide better convergence than existing classical approaches, which have trouble with high-dimensional restrictions.
- The novel inspiration for CDO and BTO comes from a distinctive and significant historical and real-world occurrence that provides new insights into optimization methods.
- The designs of CDO and BTO allow for effective exploration of the solution space domain of any issue by

avoiding local optima and striking a balance between exploration and exploitation tactics.

- CDO is a strong and effective method for resolving challenging optimization issues.
- CDO employs random tuning parameter values, which correspond to the precise values of the alpha, beta, and gamma particles spreading speeds as well as the human walking pace. Additionally, it applies the logarithm to these data in order to normalize them.
- To avoid local minima, BTO also uses random variables in its process.

However, they are facing various problems, such as tripping in local minima, slow convergence, and bad local searching ability [2,4,5,8,9,14]. For these reasons, we are motivated to propose new optimization algorithm that mitigate these problems. This optimizer called “Bombardier Beetle Optimizer (BBO)”, which inspired by the defense mechanism of Bombardier Beetle against the predators.

The rest of the paper is listed as follows. The Bombardier Beetle behavior, its chemical reactions and Bombardier Beetle Optimizer are discussed in Sec. 2. Experimental and outcomes of the proposed method are presented in Sec.3. We conclude the outcomes in Sec.4.

II. REVIEW ON BOMBARDIER BEETLE

The “Carabidae family” of bombardier beetles has a special and potent chemical defense system. It repels predators with a scorching, toxic chemical spray when it feels threatened. This reaction occurs in a specialized reaction chamber inside its abdomen and includes a well-regulated enzymatic mechanism [15-21]. The Bombardier Beetle, while activating its chemical defense system is depicted in fig 1. (a). The abdomen and specialized reaction chambers of the Bombardier Beetle is depicted in fig 1. (b).

A. Chemical Reactions

In its abdomen, the insect stores two distinct chemicals, which are as follows [15-20]:

- Hydroquinone ($C_6H_4(OH)_2$)
- Hydrogen peroxide (H_2O_2)

These substances are combined in a reaction chamber with enzymes such as catalase and peroxidase when the beetle is in danger. A hot spray is ejected as a result of an exothermic reaction that is triggered by it. The following chemical reactions are produced as a result of chemical reactions [17]:

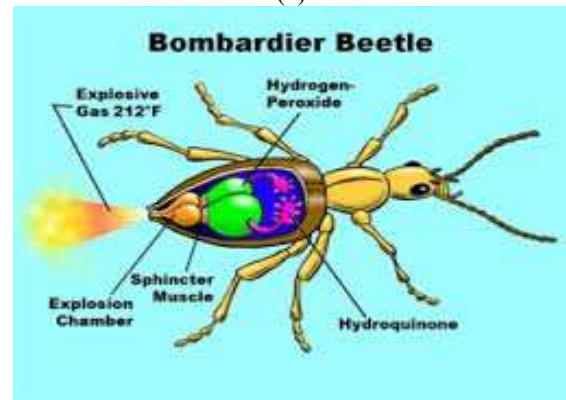
1. Hydrogen peroxide decomposition:
 - $2 H_2O_2 \rightarrow 2 H_2O + O_2$ (catalyzed by catalase)
2. Hydroquinone oxidation:
 - $C_6H_4(OH)_2 + O_2 \rightarrow C_6H_4O_2$ (quinone) + H_2O (catalyzed by peroxidase)

The outcome of this reaction can be summarized as follows, the mixture is heated to over $100^\circ C$ as a result of the reaction's

quick release of gas and heat. The toxic mixture is forced through a nozzle in bursts by the pressure. Predators are efficiently deterred by the expelled spray, which comprises hot water vapor, oxygen, and irritating substances like p-benzoquinones. Table 1 shows the components of a chemical reaction and the outcome [17].



(a)



(b)

Fig 1. The Bombardier Beetle, while activating its chemical defense system is depicted in (a). The abdomen and specialized reaction chambers of the Bombardier Beetle is depicted in section (b).

TABLE I. COMPONENTS OF CHEMICAL REACTION AND THE OUTCOME [17]

Component	Function
Hydroquinone	Fuel, oxidized to form quinones
Hydrogen peroxide	Oxidizer, decomposes to release O_2
Catalase/Peroxidase	Enzymes that catalyze reactions
p-Benzoquinone	Irritant and toxic compound
Outcome	Hot, noxious spray (defense mechanism)

B. The Bombardier Beetle Optimizer

In this paper, the prior information of defense mechanism of the Bombardier Beetle motivated us to mimics it as optimization algorithm, called Bombardier Beetle Optimizer (BBO). The algorithm steps are organized in the following points.

1) Initialization of BBO:

This algorithm proposes that there is a set of Bombardier Beetle searching a search space of a problem. Depending on where it falls in the problem-solving space, each one produces a value assignment for the decision variables. Thus, each Bombardier Beetle is a possible solution that can be represented mathematically as a vector. These vectors can be mathematically converted as a matrix as depicted in the Fig 2.

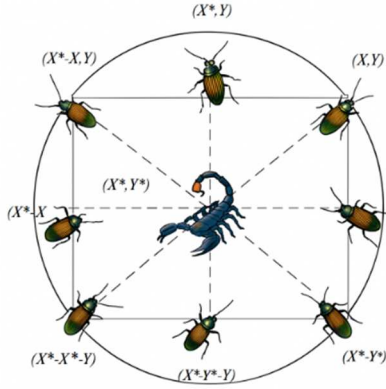


Fig 2. Positions of Bombardier Beetle and predator

The initialization of each position of Bombardier Beetle on search space should not exceed the upper bound and not less than the lower bound of the problem-solving space. The mathematical equation of initialization can be formularized as follows:

$$x_{i,d} = lb_d + r \cdot (ub_d - lb_d) \quad (1)$$

2) Defense mechanism of Bombardier Beetle

In this section, we simulate the defense mechanism of the Bombardier Beetle against the predators, which is the exploration phase of the algorithm. The area of defense is a form of circle, which it triggers a toxic chemical spray when the area of the predator intersects the area of the Bombardier Beetle. Based on that, we have three possible situations as follows:

- The predator is away from the Bombardier Beetle, which means that there is no overlap or intersection area between the Bombardier Beetle and the predator. In this case, the Bombardier Beetle will not trigger the toxic chemical spray. These areas are depicted in the Fig 3.
- The rest two cases are as follows:
 - a. The area of the predator intersects the area of the Bombardier Beetle, which is depicted in Fig. 4 (a).
 - b. The area of the predator overlaps the area of the Bombardier Beetle, which is depicted in Fig. 4 (b).

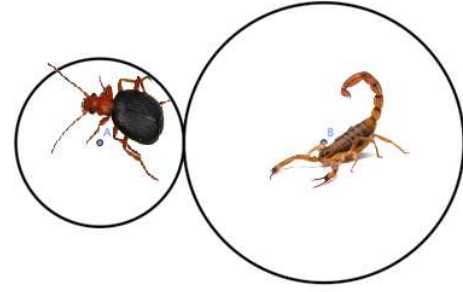


Fig 3. The predator is away from the Bombardier Beetle, which is no intersection area between them (area equal to zero).

In these two cases, the Bombardier Beetle will activate the defense mechanism by triggering the toxic chemical spray to the predator. These two cases are depicted in figure 4 (a), and (b).

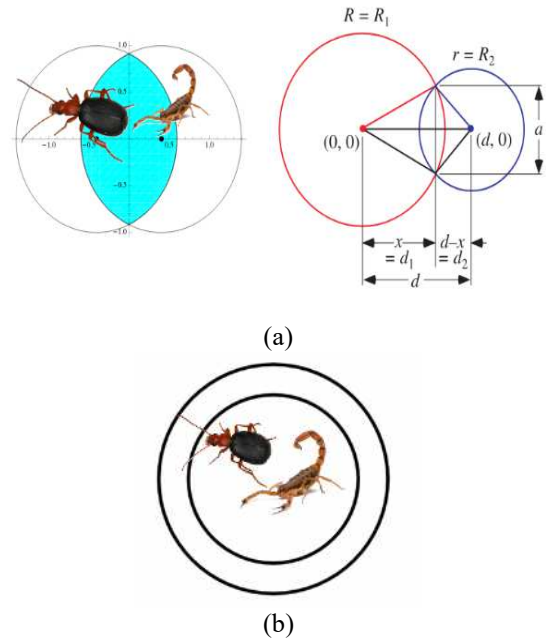


Fig 4. The area of predator intersects with the area of Bombardier Beetle is depicted in (a). The area of the predator overlaps with the area of Bombardier Beetle is depicted in (b).

Mathematically speaking, the following equation is used to calculate the intersection area of two circles, which is the area of spraying the toxic chemical spray.

$$A = A(R, d_1) + A(r, d_2) \quad (2)$$

$$= r^2 \cos^{-1} \left(\frac{d^2 + r^2 - R^2}{2dr} \right) + R^2 \cos^{-1} \left(\frac{d^2 + R^2 - r^2}{2dR} \right) - \frac{1}{2} \sqrt{(-d + r + R) \times (d + r - R) \times (d - r + R) \times (d + r + R)}$$

where,

- R is the radius of circle number one, which is the area of the Bombardier Beetle;
- r is the radius of circle number two, which is the area of the predator;
- d is the distance between the centers of the circles.

As we aforementioned, mathematically speaking, we have a set of possible solutions, which are as follows:

1. If d is greater than the radius of circles number one and two, this means that the circles do not overlap, which the intersection area is equal to zero.
2. If d is less than or equal to the absolute value of the subtraction between the radius of circles number one and two, this means that one circle is fully inside the other.

The Bombardier Beetle selects the position of a predator to activate the defense mechanism. Depends on the modeling of Bombardier Beetle movement and the intersection area between Bombardier Beetle and the predator, a new position for each individual of the population should be calculated using Eq. (3). Hence, the procedure will replace the prior location if the new objective function value is better than the old one using Eq. (4).

$$x_{i,d}^{PBB} = \frac{(x_{i,d} + (SelectePredator_{i,d} \times CIA \times CR \cdot x_{i,d}))}{Spray} \quad (3)$$

$$, i = 1, 2, \dots, N, \text{ and } d = 1, 2, \dots, m$$

$$X_i = \begin{cases} X_i^{P1}, & F_i^{Obj} < F_i \\ X_i, & \text{else} \end{cases} \quad (4)$$

where,

- $x_{i,d}^{PBB}$ is the new suggested position of i^{th} Bombardier Beetle in the d^{th} dimension;
- $SelectePredator_{i,d}$ denotes the position while the d^{th} is the dimension of the predator;
- CIA is the circle intersection area between the Bombardier Beetle position and the predator position. This value can be calculated using Eq. (2). There are three possible solutions, first there is no intersection between the two areas, which means the area is zero. Second, there is an intersection point between two areas. Third, the area between the Bombardier Beetle position and the predator position is overlapped;
- CR is the chemical reaction of toxic spray, which contains the following components:
 - hot water vapor, which is a value of 100. This value is the boiling point of the water;
 - oxygen value is a random value between [0, 1];

- irritating substances like p-benzoquinone are a random value between [0, 1].
- Spray is the exponential equation of spreading the chemical reaction of toxic spray, which can be calculated using Eq. (5).

$$Spray = \left(chaos \times 2.7^{\left(100 \left(\frac{iteration}{Max_iterations} \right) \right)} \right) \quad (5)$$

where,

- chaos is used to simulate the real chaos of spraying the toxic spray over the problem-solving space domain, which includes the sinusoidal map, Chebyshev map, circle map, sine map, Gauss/mouse map, tent map, and iterative map [22]. These chaos maps can give an approximate value for the algorithm; however, we use sine map.
 - Iteration is the iteration value;
 - Max iteration is the maximum iteration of procedure.
- N is the number of Bombardier Beetle;
- m is the number of decision variables.
- F_i^{Obj} is the objective function value.

3) Newton's Explanation of Insect Lift, the Escape Phase of the Algorithm

Newton's second and third laws of motion, which serve as the foundation for his explanation of insect lift [23]:

- The second Law of Newton is:
 - Force is equal to mass multiplied by acceleration.
- The third Law of Newton is:
 - Action is equal to reaction.

An upward reactive force, or lift, is produced when an insect flaps its wings, accelerating air downward as depicted in Fig.5 in which the bombardier beetle's wings generate lift to escape and fly away from the predator. The empirical lift equation for flapping flight can be formulated in equation (6) [23], which is the exploitation phase of the algorithm. This is similar to airplane aerodynamics but modified for unsteady flapping motion. The full procedure of Bombardier Beetle Optimizer (BBO) is listed in Algorithm 1.

$$L = LC * 0.5 * \rho * V^2 * A \quad (6)$$

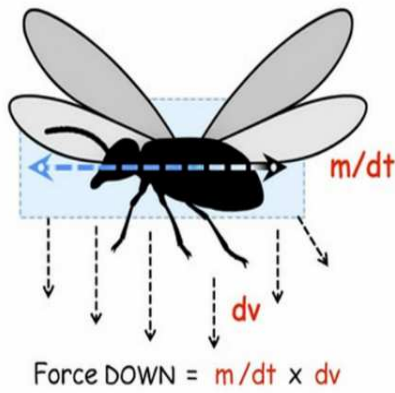


Fig 5. Flapping flight, an upward reactive force of the bombardier Beetle

where,

- LC is the lift coefficient (experimentally determined for insect wing motion);
- ρ is air density;
- V is the wing's relative velocity;
- A is the effective wing area.

Algorithm 1. Pseudocode of BBO.

Start BBO.

1. Enter the variables, restrictions, and objective function that make up the issue.
2. Set the BBO maximum number of iterations (Max_Iter), and population size (N).
3. Using Equation (1), generate the initial population matrix
4. Evaluate the objective function.
5. For $t = 1$ to Max_Iter:
6. For $i = 1$ to N :
 7. Section # 1: defense mechanism (exploration)
 - Calculated the intersection area between Bombardier Beetle and predator locations using Equation (2).
 - Apply spray defense mechanism using Equation (5).
 - Calculate the new position of the population and update the solution using Equation (3).
 8. Section #2: escape (insect lift and fly away) (exploitation)
 - Calculate the lift equation of Bombardier Beetle using Equation (6).
 - Calculate the new position of Bombardier Beetle using the following Equation:

$$\frac{L \times (ub_d - lb_d)}{t}$$
 - Calculate the new position of the population and update the solution using Equation (3). Update fitness value using Equation (4).

9. end
 10. Record the best candidate solution so far.
 11. end
 12. Output the optimal solution of the objective function.
- End BBO.

III. EXPERIMENTAL AND RESULT

In this section, CDO, SSO, GSA, PSO, BTO, GWO along with the proposed BBO are coded in “MATLAB R2023a” and run on Intel core i5 CPU with 6 GB RAM, operating Windows 11. For every test of every algorithm, a total evaluation is repeated ten times runs to ensure the correctness and conversions of every algorithm. The performance of the proposed “Bombardier Beetle Optimizer (BBO)” is estimated using the well-known six benchmark functions of “Congress on Evolutionary Computation (CEC) 2017” [24] along with all other algorithms. These test bed suites are minimization problems, which the best fitness (optimal value) of functions from 1 to 6 are zeros. The quality of results takes into account three main things for a qualitative test, which are mean (μ), standard deviation (σ), and best fitness (optimal value). The best fitness values are recorded at the end of each iteration for every algorithm. The proposed algorithm, BBO, and the other algorithms have various parameters that are configured at the beginning of the procedures. These parameters are listed in Table 2. These algorithms are compared using the following statistics:

- Mean (μ): The following formula can be used to determine the average fitness values after a certain number of generations T:

$$Mean(\mu) = \frac{\sum_{t=1}^n (f_t)}{T}, \quad (7)$$

- Standard deviation (σ): Eq. (8) can be used to determine the standard deviation between populations after a certain number of generations T. This component represents the variation in the objective model's values.

$$Std(\sigma) = \sqrt{\frac{t}{T-1} \sum_{t=1}^n (f_t - mean)^2}, \quad (8)$$

- Optimal value (best fitness): The following formula can be used to get the minimum fitness value after a certain number of generations T:

$$Best = \min_{1 \leq i \leq T} f_i, \quad (9)$$

TABLE II. PARAMETERS OF THE ALGORITHMS

Parameters	Value
SSO	
Damping factor of velocity (D)	Rand (0, 1)
pH	Rand (7, 14)
Temperature	Rand (35.5, 38.5)
Size of population (swarm size)	30
Numbers of iterations/generations	1000
GSA	
α	20
G_0	1
Size of population	30
Numbers of iterations/generations	1000
CDO	
S_γ is the speed of gamma	Rand (1, 300,000) km/s
S_β is the speed of beta	Rand (1, 270,000) km/s
S_α is the speed of alpha	Rand (1, 16,000) km/s
r is the radius of radiation's propagation	Rand (0, 1)
Size of population	30
Numbers of iterations/generations	1000
BTO	
CUG	$6.67 \times 10^{-11} \text{Nm}^2 \text{Kg}^{-2}$
$area_{\min}$	$\log(500000)$
$area_{\max}$	$\log(1,510,000)$
Size of population	30
Numbers of generations	1000
PSO	
C_1	2
C_2	2
Size of population	30
Numbers of generations	1000
GWO	
Size of population	30
Numbers of iterations/generations	1000
BBO	
Distance between Bombardier Beetle and predator (d)	Rand (0, 1)
r1 and r2 radius of circle 1 and circle 2	Rand (0, 1)
Typical insect wing lift coefficient (CL)	Rand (0, 1)
Air density (ρ)	Rand (0, 1)
Wing area (A)	Rand (0, 1)
Wing velocity (V)	Rand (0, 1)
hot water vapor of toxic spray	100
oxygen value	Rand (0, 1)
p-benzoquinones	Rand (0, 1)

From Table 3 of evaluating the first six functions of CEC2017 test suites, we can notice that the best fitness has a highlighted background. Ten-times runs are conducted to evaluate the optimization algorithms. Statistically speaking, BBO is the best in optimizing the functions from 1 to 6 of CEC2017. BBO gets the exact optimal value of these benchmark functions, which is superior and faster than the other algorithms in reaching the optimal value, especially in solving the rugged and noisy problems, such as problems from one to four. This is very clear

in the convergence rates of the BBO which are depicted in fig. 6.

IV. CONCLUSION

In this article, a new optimization algorithm, called “Bombardier Beetle Optimizer (BBO)” has been proposed. The main principles of this approach is inspired by the defense mechanism of Bombardier Beetle against the predators, which the Bombardier Beetle triggers a toxic chemical spray when it feels threatened. Six well-known test problems of CEC2017 have been chosen to compare the BBO against other well-known algorithms, such as PSO, SSO, GSA, GWO, BTO, and CDO. These approaches are compared using methodology, known as qualitative tests in which the mean (μ), standard deviation (σ), and optimal value (best fitness) are computed for this test. From the results, we can notice that the proposed BBO has merit in exploration mechanisms in solving the narrow and wide domains of these test bed problems. This is very clear in the table 3 of results. We can conclude that the BBO can be considered as a more efficient feasible alternative algorithm since it outperformed PSO, SSO, GSA, GWO, BTO, and CDO in most test bed problems of the rugged and noisy domain of search space.

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TABLE III. NUMERICAL AND STATISTICAL RESULTS OF TEST SUITES OF BBO, GWO, BTO, PSO, GSA, SSO, AND CDO

		CDO	SSO	GSA	PSO	BTO	GWO	BBO
F1	Best	2.29E-262	7.58E-228	1.01E-16	2.60E-10	0.00E+00	6.73E-60	0.00E+00
	Mean	2806.973	146.1002	211.3284	546.5455	6.237578	385.47	75.05441
	Worst	7.12E+04	64754.96	6.06E+04	7.86E+04	673.4686	7.89E+04	5.63E+04
	Std	9218.187	2452.828	2725.425	4805.066	43.35072	3768.091615	1838.36768
	Rank	2	3	5	6	1	4	1
F2	Best	2.79E-135	1.24E-129	7.53E-08	6.52E-06	3.1371e-310	5.70E-35	0.00E+00
	Mean	2.14E+10	3.78E+08	11333683	6.9E+09	0.402896	8.82E+09	1.88E+09
	Worst	1.96E+13	3.78E+11	1.13E+10	6.90E+12	21.98549	8.82E+12	1.87E+12
	Std	6.18E+11	1.2E+10	3.58E+08	2.18E+11	2.084922	2.78957E+11	59331968562
	Rank	3	4	6	7	2	5	1
F3	Best	1.83E-226	5.68E-111	5.79E+02	13.89951	0.00E+00	7.54E-15	0.00E+00
	Mean	85807.05	486.0306	1025.112	1474.157	307.9632	1706.965	205.8413
	Worst	2.05E+05	76138.58	1.03E+05	8.60E+04	14244.81	1.76E+05	1.55E+05
	Std	100057.6	4991.478	4559.733	6412.374	1536.95	9973.189196	5093.214413
	Rank	2	3	6	5	1	4	1
F4	Best	1.52E-126	1.14E-90	2.4705831	0.6105174	2.39E-303	1.58E-15	0.00E+00
	Mean	11.00552	0.862714	3.718571	3.157701	7.365007	1.450779	0.167414
	Worst	9.18E+01	87.73271	8.54E+01	8.92E+01	70.23451	8.68E+01	9.00E+01
	Std	23.72986	6.270643	5.20553	7.254041	18.39352	7.920057621	3.286272631
	Rank	3	4	7	6	2	5	1
F5	Best	27.239300	28.084450	26.834915	22.63784	8.91678572	28.7275415	0.00E+00
	Mean	7459448	679417.8	284805.2	482349.6	38370.17	666412.5	348228.7

F6	Worst	1.72E+08	2.46E+08	2.40E+08	2.41E+08	15543596	2.61E+08	3.22E+08
	Std	26860037	10241293	7657691	8810990	658704.6	10597955.6	10224107.88
	Rank	5	6	4	3	2	7	1
	Best	7.5	4.6204259	297.666	2.72E-09	0.46577979	0.75328863	0.00E+00
	Mean	3585.788	111.1483	495.7094	567.6743	408.026	320.0206	97.522
	Worst	6.34E+04	57081.7	7.55E+04	7.45E+04	9572.447	6.43E+04	7.07E+04
	Std	11301.18	1959.556	3336.817	5015.746	1583.102	3318.004931	2335.554615
	Rank	6	5	7	2	3	4	1

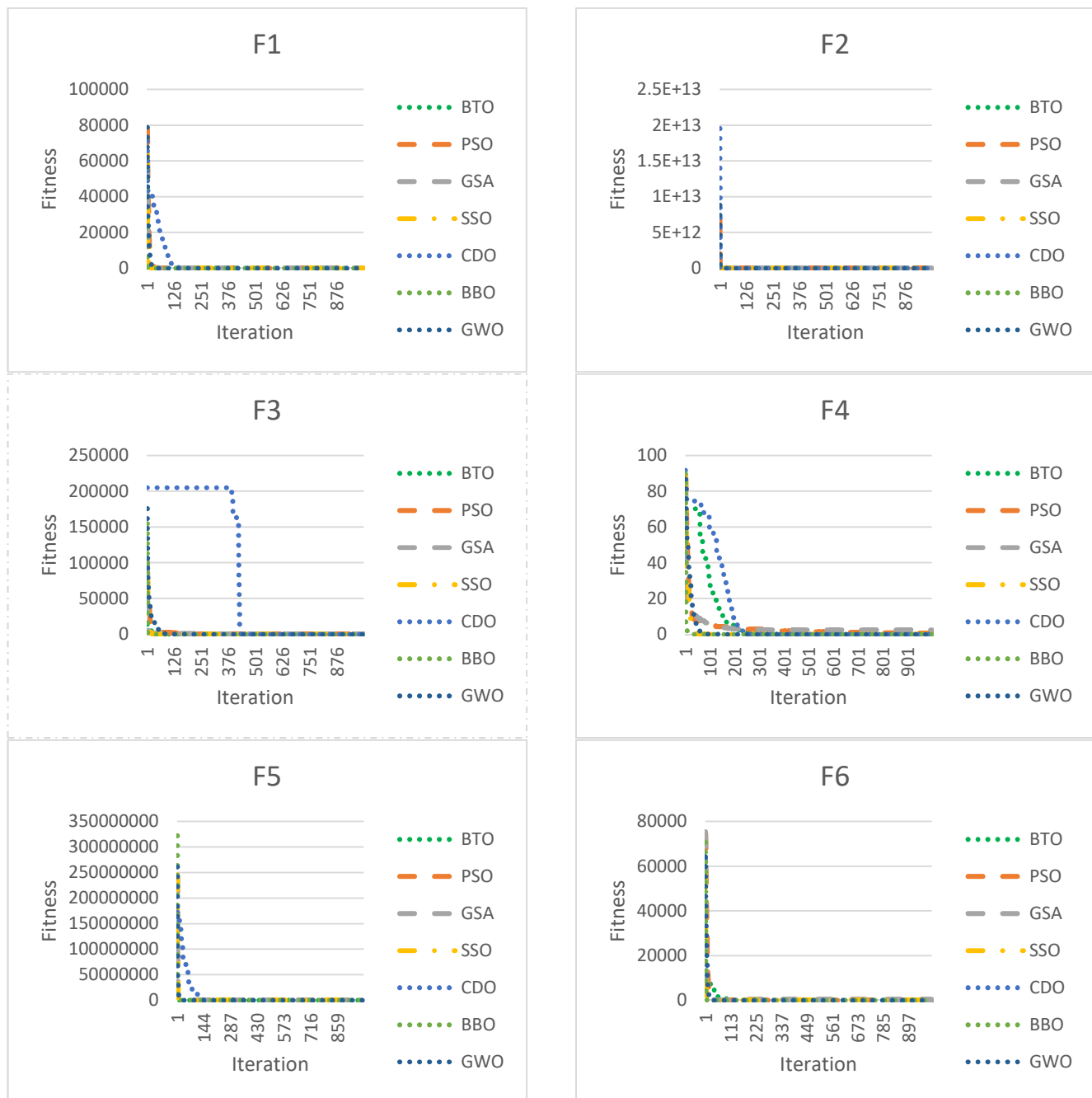


Fig 6. Convergence rates of the algorithms in solving the benchmark functions from f1 to f6 of CEC2017